AQUASHIELD CPS — PRESENTATION GUIDE

Finding & Fixing Cyber Attacks on Clean Water Plants (In Plain English)

Sustainathon 2024 • VIT Vellore
UN SDG 6: Clean Water • Zero Jargon Presentation Dossier

> **The Official Problem Statement (What the judges asked):**
> *"Detecting an anomaly does not necessarily reveal whether its origin is a sensor, communication channel, controller, firmware component or physical actuator. Cybersecurity systems can identify abnormal behaviour without providing sufficient information about where an attack originated."*

1. The Problem in Plain English (The Human Body Analogy)

Think of a modern clean water recycling plant like the human body: **sensors** are the eyes, **network cables** are the nerves, the **controller** is the brain, **software** is the instincts, and **water valves/pumps** are the muscles.

The Big Issue: Today's security tools only sound an alarm saying *"Something feels wrong!"*. They cannot tell if a thermometer is lying, a cable is cut, the computer was hacked, or a valve is jammed. Because operators don't know what broke, they panic, shut down the entire plant, and dump over **150,000 Liters of clean drinking water** into the drain!

The AquaShield Solution: Hackers can forge digital computer numbers, but **they cannot break the laws of physics**. AquaShield automatically compares incoming water vs outgoing water. In less than **38 milliseconds**, it finds the exact broken part, turns on a software backup, and fixes the plant **100% automatically without dumping any water**.

2. Live Running Dashboard (Visual Proof)



Figure 1: AquaShield CPS live demo — Attack launched, red laser locks onto the sensor, isolating the problem in 38ms while clean drinking water keeps flowing.

3. The 5 Layers (Explained in Everyday Language)

The competition problem specifically listed 5 places where an attack can hide. Here is how AquaShield tests each one:

Layer (Analogy)	What Part It Is	What the Hacker Did	How AquaShield Solves It Automatically
1. Sensor *(The Eyes)*	Water Level Sensor on Tank 2	**Fake Reading:** Froze the sensor at 32%, making the tank look empty while it's actually filling up.	**Virtual Sensor Backup:** Math proves water was flowing in, so level had to rise. We mute the lying sensor and calculate real water level using simple math.
2. Network Cable *(The Nerves)*	Modbus Network Wire	**Cable Jam:** Delayed messages by 385 milliseconds and resent old messages.	**Backup Wire Switch:** Physical water levels were normal, only cable timestamps were late. Automatically switches messages to backup wireless channel in 38ms.
3. Computer Brain *(The Brain)*	Factory Control Computer (PLC)	**Dangerous Command:** Hacked the computer to push water pressure to 8.5 bar to burst pipes.	**Safety Limit Clamp:** System has hardcoded rule: pipes burst at 5 bar. Automatically clamps pressure back down to a safe 3.2 bar.
4. Software Code *(The Instincts)*	Pump Motor Software Code	**Bad Code (Stuxnet):** Code shakes the motor to break it, while lying to the screen that it's normal.	**Digital Fingerprint Check:** Checks software code fingerprint. Mismatch caught! Automatically erases bad code and reloads clean original code.
5. Moving Parts *(The Muscles)*	Emergency Release Valve	**Stuck Metal Valve:** Solenoid valve physically jammed open at 100%, leaking water.	**Backup Pipe Diverter:** Computer said CLOSE, but water was leaking. Automatically diverts flow into a backup pipe and stops leak immediately.

4. Key App Features (No Jargon)

A. Clean Water Plant (SCADA Twin)

Full-width view of 4 water treatment tanks (Raw Water $\to$ Filter $\to$ Disinfection $\to$ Pure Drinking Water) with real-time water levels and rotating pump turbines.

B. Autonomous AI Mode vs Manual

Click one button and watch the system find the attack and fix it **100% automatically in under 3 seconds**. Switch to Manual mode to inspect each step slowly.

C. The 5-Layer Stack (Below Plant)

5 clean cards across the screen. When an attack happens, **only the broken card turns red**, while the other 4 cards prove they are safe and green.

D. Old Tools vs. AquaShield

A direct comparison: Old tools panic, shut the plant down, and dump 150,000L of water. AquaShield fixes it in 38ms with **0 seconds of downtime** and zero water wasted.

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5. How to Run the Demo (Simple Checklist)

1. Open Terminal on your Mac and run: `cd /Users/pradyun/Desktop/SUSTAINATHON && npm run dev`
2. Open Google Chrome to: **http://localhost:5173**
3. Press **Cmd + Shift + F** (or F11) on Mac to go full-screen.
4. Keep laptop volume at ~40% so judges hear the cyber beeps and victory chimes.
5. Make sure the mode is set to **Autonomous AI (Automatic)**.

6. The 2-Minute Winning Pitch Script (Rehearse This!)

STEP 1: The Hook & The Problem **0:00 - 0:30**

"Good morning judges! Our project is AquaShield CPS, solving the Attack-Localization Problem in clean water facilities.
In water plants, standard cybersecurity alarms can tell you that something is wrong, but they cannot tell you WHERE it started — whether a sensor is lying, a network cable is delayed, the computer was hacked, software is corrupted, or a valve is jammed.
Because operators cannot tell what broke, they panic, shut down the facility, and dump hundreds of thousands of liters of clean drinking water into the drain."

STEP 2: The 1-Click Autonomous Demo (The 'Aha!' Moment) **0:30 - 1:15**

"Now watch our live water recycling plant. I will launch a cyber attack targeting the water level sensor.

(Click the big blue button: 'Launch Cyber Attack & Watch AquaShield Fix It Automatically')

Look at what happens: The alarm sounds! Old tools panic because they think the tank is drying up. But in less than 2 seconds, our laser shoots across the plant and locks onto the sensor. AquaShield proved through incoming water flow that the sensor was lying. It immediately mutes the sensor, activates our virtual math backup, and keeps the plant running completely safely!"

STEP 3: The Sustainability Punchline **1:15 - 2:00**

"A traditional system would have dumped 150,000 Liters of clean water and taken 6 hours to inspect. With AquaShield, plant availability stays at 99.4%, clean drinking water never stops flowing, and zero water is wasted.

And as you just saw, the system self-heals 100% automatically without waiting for humans. Thank you!"

7. Answers to the Top 3 Judge Questions (Simple Language)

Q1: Does a human have to go fix it, or does the computer fix it by itself?
Answer: "The computer fixes the emergency 100% automatically in less than 2 seconds using software backups so clean water is never dumped. Humans only get an email report later for routine checkups."

Q2: How is this different from normal cybersecurity tools?
Answer: "Normal tools only look at computer code. They treat water tanks as a black box. AquaShield compares computer data with the real laws of physics (water flowing in vs out) to catch lies instantly."

Q3: Why is this on software instead of real hardware?
Answer: "Real water plants handle millions of liters of high-pressure water. Our software is an industrial Digital Twin simulating real fluid physics, which is the exact same method used by Siemens and GE Digital."